

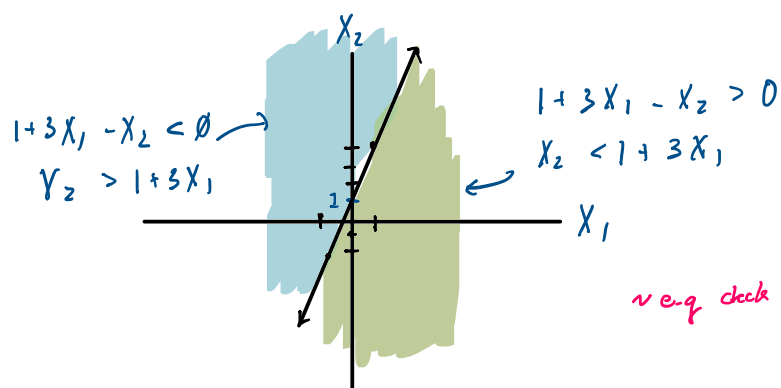
HW5 ISL 9.7

Wednesday, March 25, 2026

7:13 PM

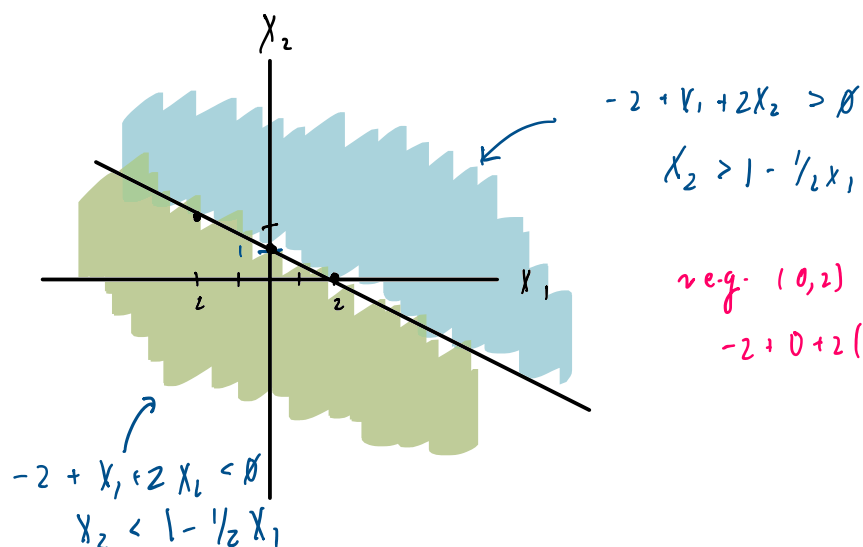
1a) hyperplane w/ 2D is just a line...

solving for X_2 : $X_2 = 1 + 3X_1$



e.g. check $(0,0) \rightarrow 1 + 3(0) - 0 = 1 > 0$

b) ~ solve for X_2 : $X_2 = 1 - \frac{1}{2}X_1$

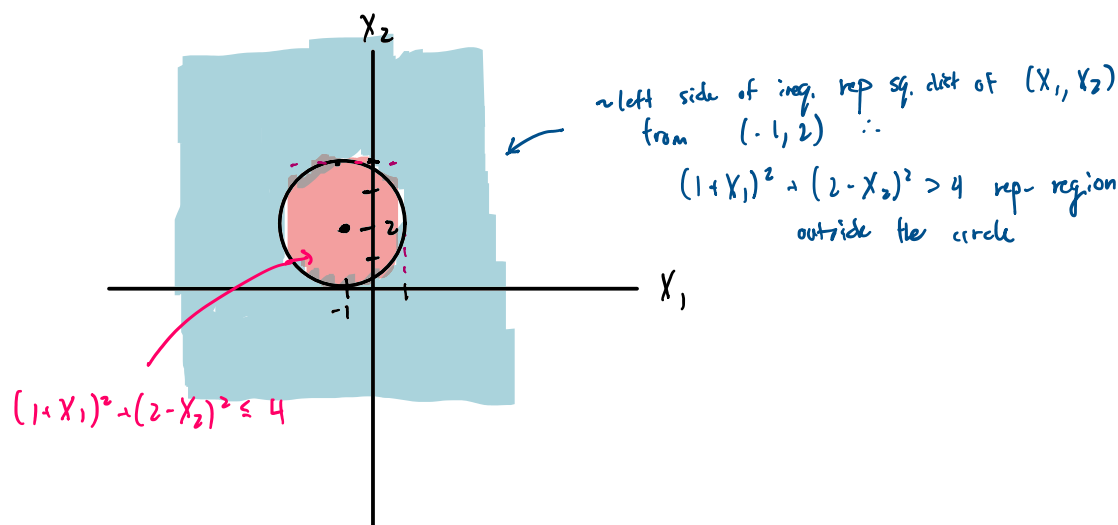


e.g. $(0,2) \rightarrow -2 + 0 + 2(2) = 2 > 0$

2a) note $(2-x)^2 = (x-2)^2 \rightarrow 4 - 4x + x^2 = x^2 - 4x + 4$

~ rewrite $(1+X_1)^2 + (2-X_2)^2 > 4$ as $(X_1+1)^2 + (X_2-2)^2 > 4$

~ circle center at $(-1, 2)$ & radius 2



~ left side of ineq. rep sq. dist of (X_1, X_2) from $(-1, 2)$ $\therefore$

$(1+X_1)^2 + (2-X_2)^2 > 4$ rep region outside the circle

b) enabled on above depiction

c) blue class: $(0,0), (2,2), (3,8)$

red class: $(-1,1)$

[can plug into $(1+X_1)^2 + (2-X_2)^2 > 4$ but visually works]

d) the decision boundary: $(1+X_1)^2 + (2-X_2)^2 = 4$

expands to $\rightarrow 1 + 2X_1 + X_1^2 + 4 - 4X_2 + X_2^2 = 4$

$\rightarrow X_1^2 + X_2^2 + 2X_1 - 4X_2 + 1 = 0$

~ if we def: $Y_1 = X_1, Y_2 = X_2, Y_3 = X_1^2, Y_4 = X_2^2$

then boundary $\rightarrow Y_3^2 - Y_4^2 + 2Y_1 - 4Y_2 + 1 = 0$

$\therefore$ we can state that the decision boundary could be linear in a higher dim. space (e.g. in terms of X_1, X_1^2, X_2, X_2^2)